

Missing input validation on `tf.ragged.constant` #55199

✓ Closed

kanghj opened on Mar 11, 2022 · edited by kanghj

Edits ...

Please make sure that this is a bug. As per our

[GitHub Policy](#),

we only address code/doc bugs, performance issues, feature requests and build/installation issues on GitHub. tag:bug_template

System information

- Have I written custom code (as opposed to using a stock example script provided in TensorFlow): yes
- OS Platform and Distribution (e.g., Linux Ubuntu 16.04):
- Mobile device (e.g. iPhone 8, Pixel 2, Samsung Galaxy) if the issue happens on mobile device:
- TensorFlow installed from (source or binary):
- TensorFlow version (use command below):2.8.0
- Python version: 3.7.12
- Bazel version (if compiling from source):
- GCC/Compiler version (if compiling from source):
- CUDA/cuDNN version: using a colab notebook
- GPU model and memory: using a colab notebook

You can collect some of this information using our environment capture [script](#)

You can also obtain the TensorFlow version with:

1. TF 1.0: `python -c "import tensorflow as tf; print(tf.GIT_VERSION, tf.VERSION)"`
2. TF 2.0: `python -c "import tensorflow as tf; print(tf.version.GIT_VERSION, tf.version.VERSION)"`

Describe the current behavior

If I pass an empty list with a large `ragged_rank` to `tf.ragged.constant`, all RAM is consumed, causing the notebook to crash.

The docs indicate that `ragged_rank` should be between 0 and the rank of `pylist`, so the large value of `ragged_rank` should be rejected

Describe the expected behavior

Some input validation should be done and an exception thrown.

[Contributing](#)

- Do you want to contribute a PR? (yes/no):
- Briefly describe your candidate solution(if contributing):

Standalone code to reproduce the issue

Provide a reproducible test case that is the bare minimum necessary to generate the problem. If possible, please share a link to Colab/Jupyter/any notebook.

The colab notebook:

<https://colab.research.google.com/drive/1OyQNTCiqHKjmHKfYbSOMVt4EfkLEgsNA?usp=sharing>

```
import tensorflow as tf
tf.ragged.constant(pylist=[], ragged_rank=8968073515812833920)
```



Other info / logs Include any logs or source code that would be helpful to diagnose the problem. If including tracebacks, please include the full traceback. Large logs and files should be attached.



kanghj added **type:bug** on Mar 11, 2022



google-mi-butler assigned sushreebarsa on Mar 11, 2022



sushreebarsa added **comp:ops** **TF 2.8** on Mar 14, 2022

sushreebarsa on Mar 14, 2022

Contributor ...

@kanghj In order to expedite the trouble-shooting process ,could you please provide the access to the colab ? Thanks!



sushreebarsa added **stat:awaiting resp...** on Mar 14, 2022

kanghj on Mar 14, 2022

Author ...

Sorry, my bad, **@sushreebarsa**. I have updated the access permissions to the colab. Could you try again?
<https://colab.research.google.com/drive/1OyQNTCiqHKjmHKfYbSOMVt4EfkLEgsNA?usp=sharing>



tensorflowbutler removed **stat:awaiting resp...** on Mar 17, 2022

sushreebarsa on Mar 17, 2022

Contributor ...

@gadagashwini I was able to replicate the issue on colab using TF v2.8.0 and tf-nightly , please find the gist [here](#) for reference.Thanks!

-  **sushreebarsa** assigned [gadagashwini](#) and unassigned [sushreebarsa](#) on Mar 17, 2022
-  **gadagashwini** assigned [sachinprasadhs](#) and unassigned [gadagashwini](#) on Mar 23, 2022
-  **rthadur** mentioned this on Apr 10, 2022
-  [Fix issue with tf.ragged.constant when empty pylist is provided #55567](#)
-   **rthadur** added a commit that references this issue on Apr 10, 2022
- Update ragged_factory_ops.py  Verified 94b9b4c
-  **sachinprasadhs** assigned [rthadur](#) and unassigned [sachinprasadhs](#) on Apr 14, 2022
-   **copybara-service** closed this as [completed](#) in [bd4d558](#) on Apr 16, 2022

 google-ml-butler bot on Apr 16, 2022

Are you satisfied with the resolution of your issue?

[Yes](#)

[No](#)

-  **tensorflow-jenkins** added 4 commits that reference this issue on May 4, 2022

Prevent denial of service in `tf.ragged.constant` 

 a2e31c2

Prevent denial of service in `tf.ragged.constant` 

 1ca14da

Prevent denial of service in `tf.ragged.constant` 

 e264676

Prevent denial of service in `tf.ragged.constant` 

 78ea939

[Sign up for free](#) to join this conversation on GitHub. Already have an account? [Sign in to comment](#)

Assignees

 **rthadur**

Labels

[TF 2.8](#) [comp:ops](#) [type:bug](#)

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

 Code with agent mode

▼

 **Fix issue with tf.ragged.constant when empty pylist is provided**
tensorflow/tensorflow

Participants



+1